

Data Structures *Linked List*

Linked List

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Data Structure is...

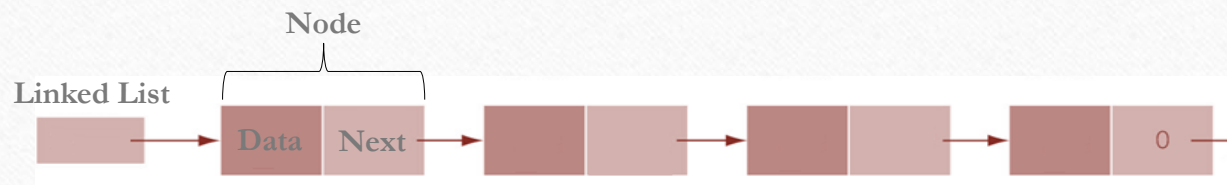
A ***data structure*** is a way of organizing, storing, and performing operations on data.

Operations performed on a data structure include:

- Accessing or updating stored data
- Searching for specific data
- Inserting new data and removing data

Linked List...

A *linked list* is a data structure that stores an ordered list of items in nodes, where each node stores data and has a pointer to the next node.



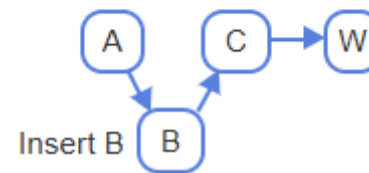
Why choose a linked list?

array:

A	B	C	W
0	1	2	3

Insert B

linked list:



Implementing a Linked List

Node:

- **Data** – Each Link of a linked list can store a data called an element.
- **Next** – Each Link of a linked list contain a link to next link called Next.
- **LinkedList** – A LinkedList contains the connection link to the first Link called First.

Basic Operations:

- **Insertion**
- **Deletion**
- **Display**
- **Search**
- **Delete**

Thinking Recursively with Linked List

- Is there a subtask that is similar to the original task?

Summary

- When implementing a linked list, we need to define the list, node, and iterator classes.
- A list node contains pointers to the next and previous nodes.
- A list object contains pointers to the first and last nodes.
- An iterator contains a pointer to the current node and to the list that contains it.
- List nodes are allocated on the free store, using the new operator.
- When a list node is erased, it is recycled to the free store with the delete operator.
- Implementing operations that modify a linked list is challenging—you need to make sure that you update all node pointers correctly.

Questions



Misc
